

UNIVERSITY OF ZIMBABWE

MOBILE SELF-CHECKOUT WITH MASK R CNN AND ATROUS CONVOLUTION

by

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THIS RESEARCH IS SUBMITTED TO THE DEPARTMENT OF COMPUTER ENGINEERING IN PARTIAL FULFILMENT OF THE BACHELOR OF SCIENCE HONOURS DEGREE IN COMPUTER ENGINEERING

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Declaration

I, Ruth P. Kaseke, submit this project as part of the requirements for a Bachelor of Science Honors Degree in Computer Engineering at the University of Zimbabwe. I affirm that, except where acknowledged, this work is original. The content and conclusions in this document have not been, and will not be, submitted to any other university for the awarding of a degree. I consent to the head of the department or their representatives granting permission for extensive copying of this project for scholarly purposes. However, copying or publishing this project for financial gain is prohibited without my written consent.

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Approval

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Abstract

This study aims to transform the retail landscape by developing a mobile-based self-checkout application integrating Mask R-CNN and atrous convolution. This initiative aims to address the inefficiencies of traditional checkout systems while enhancing customer experience and operational efficiency for retailers. Through a comprehensive analysis of user needs, business requirements, and technological capabilities, the project seeks to design an intuitive interface facilitating real-time object detection, cart management, and secure payment processing.

The project methodology entails a case study approach, starting with data collection and analysis to understand retail item characteristics and user preferences. Leveraging machine learning techniques, particularly Mask R-CNN and atrous convolution, the project trains a model capable of accurate and efficient object detection and recognition.

The project's successes are evident in the successful integration of advanced technologies, resulting in a high-performance self-checkout application. Key achievements include the development of an intuitive user interface, precise object detection capabilities, and efficient cart management functionalities.

In conclusion, this study represents a significant advancement in retail technology, offering a scalable and efficient solution to streamline the checkout process by leveraging advanced machine learning techniques.

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Chapter 1

Introduction

In recent years, the retail industry has seen a shift towards improving the customer experience through innovative technologies. One notable advancement is the introduction of self-checkout solutions, which give users the autonomy to scan and purchase items without relying on traditional cashier assistance [3].
Self-check solutions that are conventional tend to be based on standard object recognition models and are effective in controlled environments, but they may encounter difficulties when dealing with complex product presentations, diverse settings, and complex scenes [10]. These challenges can lead to inaccuracies in object detection and item recognition and subsequently influence the overall user experience. To address these limitations, the proposed work addresses the fusion of Mask R-CNN and Atrous convolution within a mobile-based self-checkout setting.

1.1 Background of Study

Accurate object detection is fundamental to the success of self-checkout platforms, and users rely on systems that accurately identify and classify scanned products [10]. Traditional methods can struggle with occlusion, inaccurate demarcation, and adaptability to different product presentations [10]. Recognizing the need for a more sophisticated and adaptive approach, this work leverages the capabilities of Mask R-CNN, an advanced instance segmentation model known for its ability to define object boundaries accurately.
While Mask R-CNN contributes to an accurate depiction of objects, the integration of Atrous convolution improves the application's ability to understand contextual information in images. By creating gaps between filter values, atrous or dilated convolution can increase the receptive field without excessively increasing parameters. This improvement is valuable in scenarios with varying lighting conditions or complex product placement, as it provides a more refined understanding of visual context.
As mobile devices are a common feature in everyday life, this study concentrates

on adapting advanced computer vision techniques to work better with these platforms. The goal is to create a seamless, responsive self-checkout experience that meets the convenience and mobility expectations of modern consumers.

Research Gaps and Contributions
Despite advances in computer vision, the integration of Mask R-CNN with Atrous Convolution into mobile self-checkout solutions represents a new research avenue. While the existing literature provides valuable insight into the individual components, a clear research gap exists in the comprehensive combination of these technologies specifically tailored to the challenges of self-checkout environments. This research aims to address this gap by providing a comprehensive solution that optimizes object detection accuracy, context understanding, and overall performance within the confines of mobile devices. The discoveries ought to facilitate the evolution of self-check applications towards more dependable, adaptable, and user-centric solutions.

1.2 Problem Definition

The use of self-checkout solutions in retail businesses promises a transition to a more efficient and convenient shopping experience [3]. However, an examination of existing systems reveals persistent challenges that prevent seamless integration into diverse retail environments. The purpose of this problem definition is to clarify the limitations observed in modern self-checkout systems and highlight the need for innovative solutions.
The existing system shortfalls include:

1. **Object Detection Inaccuracies**
   Impact: Inaccuracies in object recognition affect the overall reliability of self-checkout transactions, leading to mispricing, misclassification, and customer dissatisfaction.

2. **Imprecise Item Segmentation:**
   Impact: Imprecise segmentation contributes to difficulties in recognizing individual items, potentially leading to errors in transaction recording and creating friction in the user experience.

3. **Limited adaptability to dynamic environments:**
   Impact: The inability to adapt to changing lighting conditions creates significant limitations that affect system performance and reliability in real-world and often unpredictable retail environments.

4. **Resource Intensity and Processing Latency:**
   Impact: Resource-intensive models can cause delays in transaction processing, reduce the efficiency promised by self-checkout systems, and cause bottlenecks during times of congestion.

Significant need for innovation:
Deficiencies identified in existing self-checkout systems highlight the urgent need for innovative solutions to address these challenges comprehensively. This problem definition lays the foundation for exploring innovative solutions such as the proposed area of research. The overall aim is to increase the efficiency, accuracy, and adaptability of self-checkout solutions to meet the dynamic needs of the modern retail industry.

1.3 Aim

Aim:
The main goal of the proposed self-checkout application is to demonstrate the feasibility and effectiveness of a mobile self-checkout system that employs Mask R-CNN with atrous convolution for accurate object detection and instance segmentation in real-world grocery checkout scenarios.

1.4 Objectives

Objectives:

1. Implement Mask R-CNN with Atrous Convolution to achieve precise object detection and segmentation.
2. Design an intuitive and easy-to-use mobile application interface that facilitates image capture, real-time object recognition, and virtual shopping cart management.

Research Questions

1. To what extent does the proposed system accurately detect and segment objects in real-world self-checkout scenarios with diverse lighting, occlusion, and object arrangements?
2. Can the system process and analyze captured images efficiently enough to ensure a smooth and responsive user experience during self-checkout?

The aims, objectives and research questions collectively strive towards the development of an innovative and practical self-checkout application that addresses existing limitations and establishes new standards for accuracy, efficiency, and user experience in the retail industry.

1.5 Methods and Instruments

Methodology:
The methodology for developing and implementing the proposed self-checkout

application, integrating Mask R-CNN with atrous convolution, involves a multi-step approach encompassing model development, dataset preparation, training, optimization, and integration into a mobile platform. The process is detailed below:

1. **Problem Definition and Application Requirements:** The problem is clearly defined, taking into account the shortcomings of existing self-checkout solutions. Accuracy benchmarks, real-time processing constraints, and mobile platform compatibility are also considered.

2. **Dataset Collection and Preparation:** A diverse dataset representative of real-world self-checkout scenarios, considering variations in lighting, item placement, and potential occlusions is curated. The dataset is also annotated with precise segmentation masks and bounding boxes for training the Mask R-CNN model.

3. **Model Development and Training:** The Mask R-CNN architecture is implemented and incorporates atrous convolution layers for enhanced contextual understanding. The model is initialized using pre-trained weights on a relevant dataset and hyper-parameters, including learning rates and dilation rates, to achieve the desired balance between accuracy and efficiency.

4. **Evaluation Metrics and Testing:** Defined evaluation metrics, including Intersection over Union (IoU), precision, recall, and F1 score, are used to assess the performance of the trained model. The dataset is split into training and testing sets to evaluate the model's generalization capability.

5. **Mobile Platform Integration:** The trained model is adapted for deployment on mobile platforms using frameworks like TensorFlow Lite or ONNX. The model size and computational requirements are optimized to ensure efficient execution on mobile devices.

6. **User Interface Design and Development:** An intuitive and user-friendly interface is designed for the self-checkout mobile application. It implements functionalities for image capture, real-time object detection, and virtual cart management.

7. **System Integration and Validation:** The trained model and the mobile application are integrated. Thorough system validation checks that ensure seamless interoperability, accurate object detection, and real-time processing on the mobile platform are conducted.

Instruments and Tools

1. ***Programming Languages:***

   - Python for model development and training
   - Flutter for mobile application development

2. ***Deep Learning Frameworks***

    - PyTorch for implementing and training the Mask R-CNN model.

This methodology and the associated instruments provide a structured approach to developing, testing, and integrating the proposed self-checkout application, ensuring a robust and efficient solution that addresses the identified challenges in existing systems.

1.6 Feasibility Study

A feasibility study is conducted to assess the viability and practicality of implementing the proposed self-checkout solution, integrating Mask R-CNN with atrous convolution and optimized for mobile platforms.

1. **Technical Feasibility:**
   Objective: Evaluate the technical aspects of implementing the proposed solution.

    - Model Implementation: The implementation of Mask R-CNN with atrous convolution is technically feasible, given the availability of mature deep learning frameworks like TensorFlow and PyTorch.
    - Mobile Platform Compatibility: The adaptation of the model for mobile platforms using TensorFlow Lite or ONNX is technically viable, ensuring compatibility with a wide range of mobile devices.

2. **Economic Feasibility:**
   Objective: Assess the economic viability of developing and implementing the proposed solution.

    - Development Costs: The development costs associated with dataset collection, model development, mobile application design, and system integration are minimal making the development of the proposed solution feasible.
    - Return on Investment (ROI): The potential benefits in terms of improved operational efficiency, reduced errors, and enhanced customer satisfaction, far outweigh the development and operational costs indicating a high ROI.

3. **Operational Feasibility:**
   Objective: Evaluate the practicality of implementing and operating the proposed application within a retail environment.

    - Ease of Integration: The application can be easily integrated into the retail landscape without much disruption to the checkout process.
    - User Acceptance: The acceptance of the mobile application by end-users, including both customers and retail staff, can be accurately determined through pilot testing and user feedback.

4. **Schedule Feasibility:**
   Objective: Assess the feasibility of completing the project within a reasonable timeframe.

   - Development Timeline: The developed timeline indicates the project can be completed within a reasonable timeframe.

Conclusion:
The feasibility study indicates that the proposed self-checkout mobile application is technically, economically, and operationally feasible. The anticipated benefits, including improved accuracy, efficiency, and user experience, align with the project's goals.

1.7 Justification and Rationale

Significance of the Project
The proposed self-checkout solution, which integrates Mask R-CNN and atrous convolution and is optimized for mobile platforms, has a significant impact on the retail industry and beyond.

1. **Technological Innovation in Retail**: The integration of advanced computer vision technology into self-checkout applications represents a technological leap forward in retail. This project will contribute to and create continuous innovation in the field of retail technology.

2. **Mobile Platform Adaptability:** The focus on mobile platform optimization is consistent with modern consumers' dependence on mobile devices. This project targets the mobile lifestyle and provides a convenient and accessible self-checkout solution.

3. **Enhanced Customer Experience:** The project aims to improve accuracy, streamlining the self-checkout process to create a more seamless and user-friendly experience for customers which can improve customer satisfaction and loyalty.

4. **Retailer Operational Efficiency:** Accurate object recognition can enhance operational efficiency, decrease transaction errors, and reduce the need for manual intervention among retailers. This optimizes processes and reduces costs.

5. **Sustainable Development Goals (SDGs) Alignment:** This project is aligned with several United Nations Sustainable Development Goals and contributes to broader social and environmental goals.

   - SDG 9: Industry, Innovation and Infrastructure: This project will foster innovation in retail and build a robust infrastructure. It will also contribute to the development and advancement of industry practices.

- SDG 12: Responsible Consumption and Production: By reducing errors in self-checkout transactions and optimizing operational efficiency, this project supports responsible consumption and production practices.
- SDG 11: Sustainable Cities and Communities: As part of the broader retail ecosystem in urban environments, the project's efficiency improvements streamline retail operations and reduce checkout congestion, which contributes to sustainable and resilient cities.

In summary, the importance of this project is that it positively affects customer experience, improves operational efficiency, contributes to innovation in the retail industry, and is consistent with key Sustainable Development Goals.

1.8 Work Plan

Project Schedule

Activity	Start Date	End Date	Duration(Days)
Requirements Gathering	27/10//2023	13/11/2023	17
Design and UI Development	14/11/2023	31/12/2024	48
Backend Development	01/01/2024	31/01/2024	31
Integration and Testing	01/02/2024	29/02/2024	29
Bug fixing and Refinement	01/03/2024	31/03/2024	31
Deployment and Launch	01/04/2024	30/04/2024	30
Approximate Duration			186

Table 1.1: Project Schedule

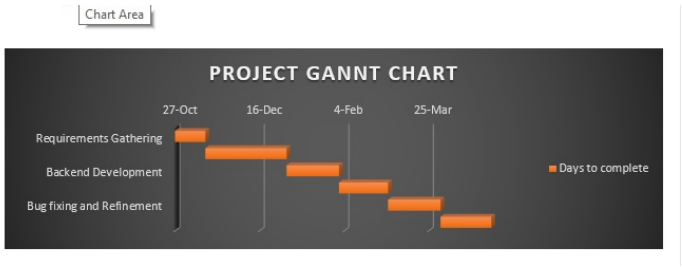


Figure 1.1: Project Gantt Chart

1.9 Conclusion

In conclusion, self-checkout solutions, including the proposed mobile application, offer significant benefits in terms of efficiency, accuracy, and customer satisfaction. The new research avenue that this research exploits seeks to provide a comprehensive solution that optimizes object detection accuracy, context understanding, and overall performance within the confines of mobile devices. The discoveries ought to facilitate the evolution of self-check applications.

Chapter 2

Literature Review

2.1 Introduction

The subsequent Literature review seeks to offer an in-depth review of the reasons for carrying out the proposed research by delving deep and analysing the factors, which led to this study. These factors include analysis of self-checkouts, various work related to self-checkouts, mask rcnn and areas where it has been used and finally summing everything with the project gap stamp.

2.2 Self-Checkout Systems

Self-checkout is a system that allows customers to complete their shopping by themselves. The use of self-checkout systems is particularly popular in supermarkets in developed countries, as it not only reduces the number of staff needed to manage the checkout but also improves productivity and customer service [19]. Because you no longer require human resources to maintain checkouts, you can increase the number of active checkouts and reduce the number of queues[19].
Traditional cash register systems primarily use barcodes. A barcode is a machine-readable visual representation of data [1]. The cash register implementation uses barcodes as unique markers to track products, thus speeding up the purchasing process. However, barcodes are difficult to read on plastic-wrapped products that do not have a three-dimensional shape, so they are not effective in introducing self-checkout systems. Barcode reading is also affected by the condition of the product's barcode. Scratches during shipping or defects in barcode printing can make the barcode unreadable by a barcode scanner. This weakness is most noticeable when used by untrained customers who are new to barcode scanning. Self-checkout is meant to speed up the purchasing process, which lengthens it. Because of these weaknesses and issues, a new alternative system replaced barcodes and it accurately reads products and is easy for customers to use. The alternative solution uses deep learning and object detection. Object detection

is a technology related to computer vision for detecting classes of objects in images [1]. Currently, several deep learning-based self-checkout systems have been implemented, such as Alibaba Group's Taocafe and Amazon's Amazon Go [6]. Both implementations use a combination of sensors, cameras, and object recognition to track customers and their purchases, allowing them to complete their purchases without waiting.

2.3 Mask RCNN

Mask R-CNN or mask region-based convolutional neural network[8] is a powerful deep-learning model for object detection and instance segmentation. Developed by[8], it builds on the success of Faster R-CNN[18] by integrating a mask prediction branch that generates a high-quality segmentation mask for each detected object[8]. This allows Mask R-CNN to not only locate an object but also understand its exact shape and boundaries, making it a valuable tool for various computer vision tasks. Mask R-CNN has attracted much attention in the field of computer vision because it can perform object detection and pixel-level segmentation simultaneously.

Technical Background

Mask R-CNN extends the Faster R-CNN [8] architecture by introducing an additional branch for segmentation mask prediction in addition to bounding box and class prediction. This model uses a Region Proposal Network (RPN) to generate candidate regions for further refinement, incorporating a pixel-to-pixel segmentation network. The Mask R-CNN formulation provides a comprehensive approach to object instance segmentation [8].

Mask R-CNN extends its RPN framework with its third branch dedicated to mask prediction. This branch receives the feature maps from the second stage and uses them to predict a binary mask for each proposed object. Masks display the segmentation of an object at a pixel level, allowing for a better understanding of object shape and boundaries.

Applications and Successes

Mask R-CNN has demonstrated its effectiveness across a range of computer vision tasks due to its ability for tasks such as object detection and instance segmentation. [8] demonstrated the model's efficacy in capturing fine-grained details of object boundaries, outperforming previous methods. It achieved good results in instance segmentation tasks, notably in the COCO dataset that the study used. Another study by [18] utilized the model to segment and analyze brain tumours in magnetic resonance images, displaying its potential in clinical applications. The model was able to successfully analyze medical images and perform organ segmentation. In 2018, [7] utilized the model for instance segmentation in video sequences, demonstrating its ability to maintain temporal consistency while accurately segmenting objects over multiple frames. This study proved the model's ability to be adapted for video analysis. [2] in 2019 applied Mask RCNN in localization and classification for Rice grain Images. The proposed Mask RCNN classified and localized the rice grain automatically.

Mask R-CNN was proposed to detect cucumber fruits by [13] in 2019. Cucumber fruits share the same colour as their leaves, and they all have elongated, narrow shapes, distinguishing them from other common fruits like apples. The detection of the fruit considering the same colour as the leaves provided a challenge for its detection, a challenge overcome by the segmentation capabilities of Mask R-CNN. To resolve traffic accident compensation issues, a segmentation algorithm for vehicle damage detection based on transfer learning was proposed by [24] using an improved Mask R-CNN. [11] in 2021 used Mask R CNN for object detection and identification of infected malaria cells using the Plasmodium Vivax dataset. The Mask R CNN model solved the instance segmentation problem and was able to detect malaria-infected cells. [12] in 2021 used Mask R CNN on a snacks dataset to establish its efficiency. The precision of the model was high with it able to accurately detect various snacks.

Conclusion

Mask R-CNN is regarded as a cornerstone of the field of instance segmentation and has demonstrated its versatility in various fields. Successful implementations in object detection, medical image analysis, video segmentation, and real-time applications in retail highlight its wide applicability and potential for further development.

2.4 Atrous Convolution

Atrous convolution, also known as dilated convolution[5], has become a powerful asset in the domain of deep learning, allowing researchers to capture larger receptive fields without significantly increasing computational complexity. This section of this paper aims to examine the theoretical foundations of atrous convolution and highlight its successful implementations in various fields.

Technical Background

Traditional CNNs are based on standard convolutions, where the filter slides over the input image with a fixed step size. This limits the size of the receptive field, i.e. the area of the input image that a particular output neuron considers when producing its output. Larger receptive fields are essential for capturing long-range dependencies and global context in images and are essential for tasks such as scene understanding, and object recognition.

Atrous convolution addresses this limitation by introducing dilation, a hyperparameter that extends the receptive field of the filter by inserting zeros between elements. This allows the filter to continue, "Referencing" without increasing the size of the filter itself, thus maintaining computational efficiency. The dilation rate determines the distance between filter elements, and the higher the rate, the larger the receptive field and the more context information is retrieved.

Applications and Successes

Atrous convolution has proven effective in a multitude of image processing and computer vision assignments. Atrous convolution has proven effective in segmenting complex objects with intricate details, such as medical images or satellite imagery. For instance, [5] achieved state-of-the-art results in retinal vessel

segmentation using atrous convolution, demonstrating its ability to capture fine-grained structures. Atrous convolution can improve image classification performance by capturing long-range dependencies within an image. Several studies, including[22] and [21], demonstrate the effectiveness of atrous convolution in enhancing classification accuracy on various datasets. The DeepLab series of models [5]leverages atrous convolution to capture contextual information over large image regions. DeepLab-v3, for instance, employs atrous spatial pyramid pooling (ASPP) to aggregate multi-scale information effectively. Remote sensing applications have benefited from atrous convolution in semantic segmentation tasks. A study by[15] applied atrous convolution to high-resolution satellite imagery, achieving improved accuracy in land cover classification. While primarily used in computer vision, atrous convolution has shown promise in NLP tasks.[24] applied dilated convolutions to text classification, demonstrating competitive performance compared to traditional convolutional approaches. [5], proposed the implementation of atrous convolution to help with the problem of reduced feature resolution due to consecutive pooling. The experiments carried out in this report support the notion of adding atrous convolution to any DCNN to extract dense feature maps and capture extensive contextual information.

Conclusion

Atrous convolution has become an important technique in deep learning, providing an efficient way to enlarge the receptive field without significantly increasing the computational complexity. Successful implementations in image segmentation, remote sensing, natural language processing and object recognition highlight its versatility and potential in various fields.

2.5 Related Work

Fruit Detection using Classic CNN

[17] employed a traditional convolutional network for fruit detection, consisting of a four-layer architecture with a filter size of (5,5). The number of filters ranged from 4 to 64, with each layer followed by a max pooling operation and fully connected layers. Their reported accuracy on a test set, encompassing 60 distinct fruit classes, reached 94 per cent.

Deep Learning based Object Detection

[11] in 2021 used Mask R CNN for object detection and identification of infected malaria cells using the Plasmodium Vivax dataset. Mask R CNN solved the instance segmentation problem and was able to detect malaria-infected cells.

Study on Mask R-CNN with Data Augmentation for Retail Product Detection

[12] in 2021 used Mask R CNN on a snacks dataset to establish its efficiency. The precision of the model was high with it able to accurately detect various snacks however it weighed on computational resources and the combination of mask r cnn and data augmentation made debugging and interpretation complex and difficult.

2.6 Project Gap Stamp

Although existing approaches excel in their specific domains, the researcher faces a unique challenge. The proposed system not only needs to detect grocery items but also accurately classify them. Additionally, it must handle scenarios where multiple instances of each grocery item are present.
From the reviewed literature, convolutional neural networks (CNNs) have been applied across various domains, yet they necessitate context-specific adjustments to facilitate tasks like image classification and object detection. Unlike traditional classification methods, CNNs do not depend on predefined mappings or markers for object detection in images. Instead, they learn classification patterns directly from data. This inherent adaptability makes CNNs an ideal choice for computer vision-based detection of packaged products within a retail setting[4].
Despite advances in computer vision, integrating Mask R-CNN with Atrous Convolution into mobile self-checkout solutions represents a new research avenue. While the existing literature provides valuable insight into the individual components, a clear research gap exists in the comprehensive combination of these technologies specifically tailored to the challenges of self-checkout environments.

2.7 Conclusion

In summary, the literature on object detection has shown that Mask R CNN and Atrous convolution are both effective methods for the task with the combination of the two providing a new research avenue that seeks to improve on the capabilities of both.

Chapter 3

Methodology

3.1 Introduction

Kaiming He, from Facebook AI Research (FAIR) [9], originally suggested the method of using Mask R-CNN for object detection. This chapter will explain how Mask R-CNN achieves precise instance segmentation and also highlight how its combination with Atrous Convolution in a hybrid approach is advantageous. The proposed methodological approach will combine sequential and parallel execution. Mask R-CNN will initially detect and coarsely locate objects, and then atrous convolution will be applied only to those detected regions in parallel for higher precision processing.

3.2 Research Design

The proposed study will utilise a case study design for the research methodology to evaluate the performance of Mask RCNN and Atrous Convolution on a mobile setup. The opted design is a single-case design where particular focus will be placed on a mobile phone, running an Android operating system.

3.2.1 Sampling Method

The opted sampling method is purposive sampling in which the researcher has selected the Grocery Store Datase, which is representative of the mobile setup this study aims to evaluate. The selected Samsung mobile device was chosen to ensure representativeness and generalizability due to the device's popularity amongst Zimbabwean users.

3.3 Model Architecture

This methodology section aims to clearly explain the functionality of Mask R CNN and its combination with Atrous Convolution. The two models will

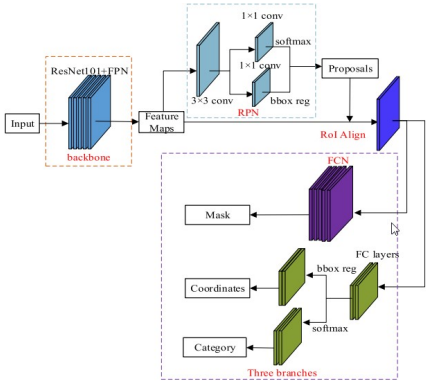


Figure 3.1: Mask RCNN

be implemented using the TensorFlow deep learning framework, which allows for model optimisation on mobile devices to accommodate the computational constraints of the device.

3.3.1 Mask R-CNN

Mask R-CNN [8] an extension of Faster R-CNN [16], improves upon its predecessor by adding an extra branch dedicated to predicting object masks. While Faster R-CNN focuses on bounding box predictions,[18] Mask R-CNN's additional mask output provides spatial details of objects, distinct from class and box predictions. Instead of having ResNet as its backbone like Faster R-CNN, Mask R-CNN employs a combination of both ResNet and feature pyramid networks (FPN). The figure above (fig 3.1) shows the structure of Mask R CNN.

Backbone model: ResNet functions as a feature extractor for the image, transforming an RGB image into a feature map that serves as input for subsequent layers.
Region Proposal Network (RPN): Using regions, and defined anchor boxes, the RPN will scan each region of the feature map and predict if an object is contained in it.
Region of Interest Classification and Bounding Box: In this stage of the algorithm, RoIs (Regions of Interest) are taken as input, and the output consists of both classification results and bounding box regression.
Segmentation Masks: In the final stage, the algorithm processes the positive RoI regions and generates masks of size 28x28 pixels, where the floating-point values represent the detected objects.
The loss function of Mask RCNN is calculated using:
$Loss(RPN) = RLoss(RPN) = RPN_ClassLoss + RPN_BBoxLoss$
Where Loss (RPN) is Loss of Region Proposal Network.
$Loss(MaskRCNN) = Loss(classlabelsprediction + Loss(Boundingboxprediction +$

$Loss(Maskprediction)$
Where Loss (Mask RCNN) is Loss of Mask RCNN.
$TotalLoss = Loss(RPN) + Loss(MaskRCNN)$

3.3.2 Mask RCNN with Atrous Convolution

Deep Convolutional Neural Networks (DCNNs) such as Mask R CNN, deployed in a fully convolutional fashion [14] are effective for semantic and instance segmentation tasks. However, the frequent use of max-pooling and striding in consecutive layers of these networks drastically decreases the spatial resolution of the resultant feature maps, typically by a factor of 32 in recent Deep Convolutional Neural Networks (DCNNs) across each direction [20]. To mitigate this, the use of atrous convolution is proposed, which is effective for semantic image segmentation [23].
Traditionally, convolutional filters have limited receptive fields (RF), affecting their ability to differentiate overlapping or occluded objects [8]. Atrous convolution addresses this by inserting "holes" (dilations) between filter elements, effectively increasing the RF without enlarging the filter size [22]. This larger RF captures broader context around potential object locations, enabling Mask R-CNN to better distinguish occluded or partially visible objects (e.g., fruits behind leaves in a supermarket checkout image).
$RFatrous = (K-1)XDilationrate + 1$
Where RFatrous is the receptive field of the atrous convolution, K is the kernel size, and the dilation rate is the factor by which the kernel is spaced.
Increasing the RF through standard convolutions requires larger filters, leading to more parameters and higher computational demands. Atrous convolution offers an alternative by achieving a similar RF expansion using dilations instead of larger filters. This can be particularly beneficial for mobile applications, where computational resources are limited.
The computational cost of a convolutional layer with kernel size KxK and dilation rate d can be approximated as:
$Cost = O(K^2xd^2xCxHxW)$
Where C is the number of input channels, H and W are the image dimensions. As d increases, the cost grows quadratically, but it remains considerably lower than increasing both K and d. This computational efficiency makes atrous convolution a valuable option for resource-constrained applications.
The larger RF of atrous convolution allows it to capture subtle details within individual objects, leading to more precise and accurate masks in Mask R-CNN's output [8]. This is particularly valuable in scenarios where accurate distinction between object boundaries is crucial, like fruit or vegetable identification in a self-checkout system.

3.4 Dataset

The research will create a dataset for validation of the algorithm to make it more suitable and applicable to the Zimbabwean retail context. The training and preliminary validation data source is from Grocery Store Dataset which is a dataset containing natural images of grocery items. The dataset includes a diverse range of object classes and covers various real-world scenarios.

3.5 Training Procedure

The Mask R-CNN and Atrous Convolution models will be trained on a high-performance computing system equipped with GPUs, a system that the Zimbabwe Centre for High-Performance Computing (ZCHPC) will provide. The models will be fine-tuned on mobile datasets to minimize the loss function, which combines classification, bounding box regression and mask prediction losses.

3.5.1 Mask R-CNN with Atrous Convolution Pseudocode

Stage 1: Region Proposal Network (RPN)

1. Input: Image (I)
2. Convolutional Layers: Apply multiple convolutional layers to extract features from I.
3. Anchor Generation: Generate a set of potential object locations (anchors) with different sizes and aspect ratios across the feature maps.
4. Objectness Score and Bounding Box Regression: For each anchor, predict its probability of being an object (objectness score) and refine its bounding box size and position (regression).
5. Non-Max Suppression: Apply non-max suppression to select a set of high-scoring and non-overlapping anchor proposals.

Stage 2: Feature Extraction and Prediction

1. Input: Anchor proposals from RPN.
2. ROI Pooling: Extract features from the convolutional feature maps for each ROI (Region of Interest) defined by the anchor proposals.
3. Shared Convolutional Layers: Apply several convolutional layers to the extracted ROI features to share computations and extract higher-level features.
4. Class Prediction: Predict the class label (e.g., apple, banana) for each ROI

5. Box Refinement: Refine the bounding box regression for each ROI
6. Atrous Convolutional Layers: For each ROI, apply one or more atrous convolutional layers to capture larger contextual information and predict a pixel-level mask.
7. Mask Prediction: Add a final classifier on the output of the atrous convolution layers to predict the binary mask for each object instance within the ROI.

Output:

1. Image with overlaid bounding boxes and masks for each detected object instance.
2. Class labels and confidence scores for each object.

3.5.2 Experimental Setup

After successful training of the model, a mobile application will be developed to integrate the model for real-time object detection. The trained model will be deployed on a selected mobile device and optimized for efficient execution considering computational constraints, memory limitations and power consumption.

3.6 Evaluation Metrics

The dataset will be split into training and test data, to establish the accuracy of the model. The efficiency of the algorithm will be measured by the average compute time (ACT) of an image as well as the mean average precision (mAP) of the algorithm. The inference time as well as memory usage on mobile devices will also be calculated to allow assessment of the model's computational cost. The algorithm will also be compared to other published works which used other methods for object detection. Particular focus will be given to the comparison of accuracy to that of [12] who achieved a 92.98 per cent accuracy for retail product detection using Mask R CNN.

3.7 Conclusion

The proposed methodology for this research aims to answer the research questions and achieve the project-set objectives. Implementing a case study methodology is anticipated to yield valuable insights into the performance of the proposed model in a mobile device setting and set the stage for future research work.

Chapter 4

Analysis and Design

4.1 Introduction

In recent years, there has been a development in automated retail systems with cutting-edge approaches in system integration, automating processes and analysing big data. In this scenario, modern deep learning techniques play a pivotal role in the design of innovative systems concerning data-driven approaches. The following chapter presents all the design phases that have defined the development of the Mobile-based self-checkout system.

4.2 Detailed Analysis of the problem domain and user requirements

The development of a mobile-based self-checkout system aims to enhance the shopping experience by reducing checkout times, minimizing human errors, and providing a seamless and efficient process. Integrating advanced image recognition techniques like Mask R-CNN (Mask Region-based Convolutional Neural Networks) and atrous convolution (also known as dilated convolution) ensures high accuracy in item detection and recognition within various environments and conditions.

The following is a list of stakeholders and their goals associated with the development of the self-checkout system.

1. Retail Customers: Primary users who desire a quick, efficient, and user-friendly checkout process.

2. Retail Store Owners and Managers: Interested in reducing operational costs, increasing customer satisfaction, and minimizing shrinkage.

3. Cashiers and Staff: Might have concerns about job security but can be redeployed to more customer-focused roles.

4. Developers and Technologists: Responsible for creating and maintaining the system, ensuring it meets technical requirements and user needs.

5. Business Analysts and UX Designers: Focused on understanding user needs and ensuring the system provides a positive user experience.

The introduction of a mobile-based self-checkout with Mask RCNN and Atrous Convolution will address challenges associated with item recognition, that is diverse product shapes, sizes and packaging, facilitate real-time processing and improve the user shopping experience. It aims to reduce checkout time and streamline the shopping process, providing a hassle-free shopping experience. Minimizing labour costs associated with manual checkout processes is also one of the project goals as well as improving accuracy in item recognition to ensure correct pricing and error reduction.

4.2.1 Functional Requirements

The system's functional requirements include the ability to extract the following information from images:

- class of each identified product
- bounding box position of each identified product
- score of each identified product

4.2.2 Non-functional Requirements

The non-functional requirements include:

- The system should process product scans and recognitions in under 2 seconds.
- Ensure real-time updates of the shopping cart.

4.3 Identification of system components and functionalities

The system contains the following functionalities:

1. User Authentication and Management which allows users to create accounts, log in securely and a password recovery option.

2. Product Scanning and Recognition which uses a mobile device's camera to scan products, uses mask rcnn for object detection and segmentation and product information retrieval.

3. Shopping Cart Management which allows users to add recognized products to their shopping cart, edit cart items and view the contents of the cart.

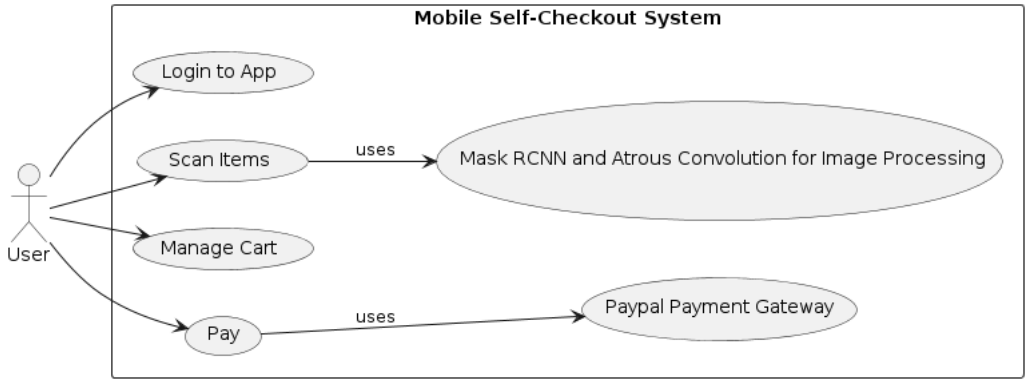


Figure 4.1: Use Case Diagram

4. Payment Processing which has an integrated payment gateway to allow users to pay for items.

Contribution to the Overall System
Each component and module of the self-checkout system plays a crucial role in delivering a seamless, efficient, and secure user experience. The User Authentication and Management module ensures that only authorized users access the system, maintaining security and personalization. Product Scanning and Recognition form the system's core, enabling accurate and quick identification of items. Shopping Cart Management and Payment Processing streamline the shopping process, ensuring users can easily manage their purchases and complete transactions securely.

4.3.1 Use-Case diagram

The diagram [fig 4.1] shows the following elements:

1. **Actors:** The user represents the person using the mobile self-checkout system.
2. **Use Cases:** The user logs into the mobile application, scans items using the mobile application, edits items in the cart and allows payment processing.
3. **Relationships:** The User interacts with all the main functionalities (Login to App, Scan Items, Manage Cart, Pay).

4.3.2 Sequence diagram

The following sequence diagrams [fig 4.2- 4.4] show the flow of messages and events that can occur as the user interacts with the mobile application.

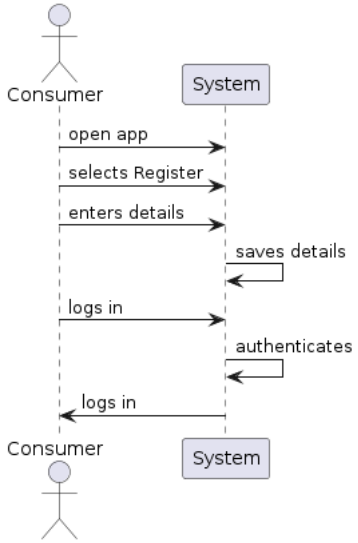


Figure 4.2: Consumer Registration

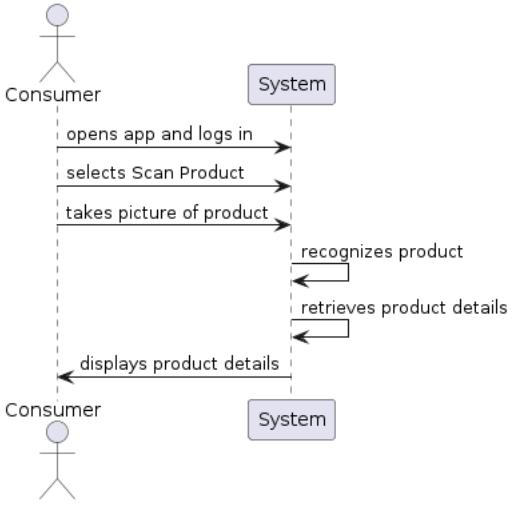


Figure 4.3: Product Scanning and Recognition

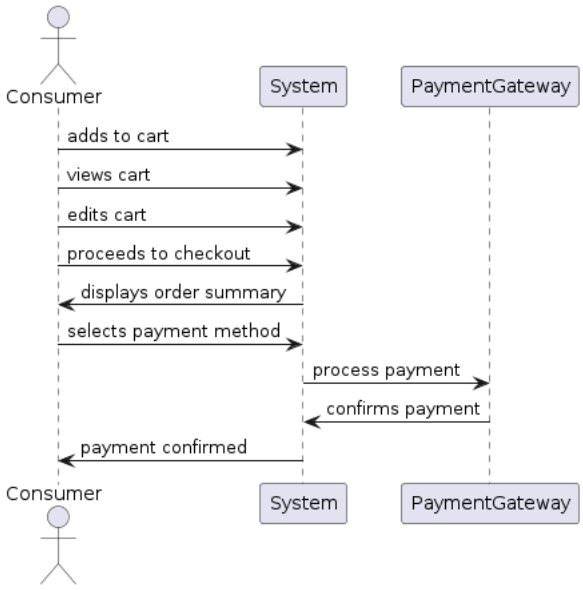


Figure 4.4: Cart Management and checkout

4.4 System Architecture and Design Considerations

4.4.1 Context Diagram and DFD Diagram

The context diagram [fig 4.5] provides a high-level overview of a system and its interactions with external entities (actors). Fig 4.6 is the DFD which provides a clear and concise visual representation of how data flows within the system. It shows how data enters the system, the processes that transform it, and how it is stored and outputs generated.

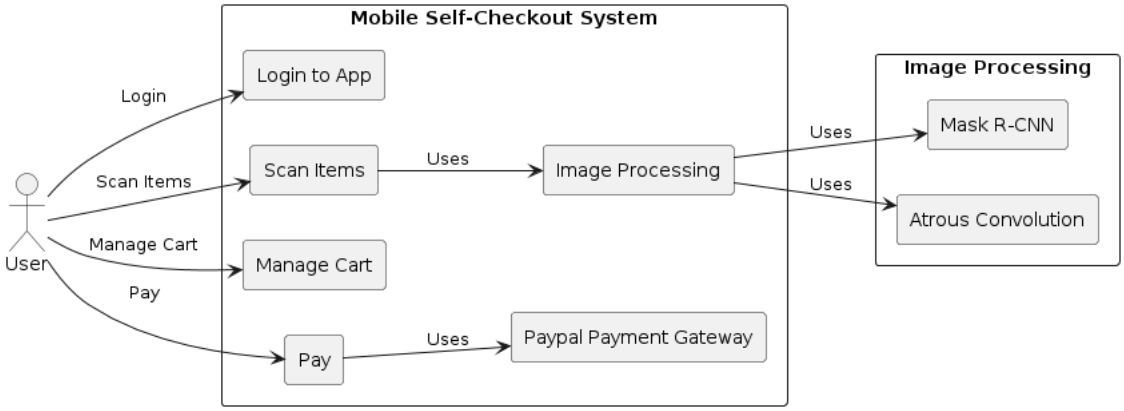


Figure 4.5: Context Diagram

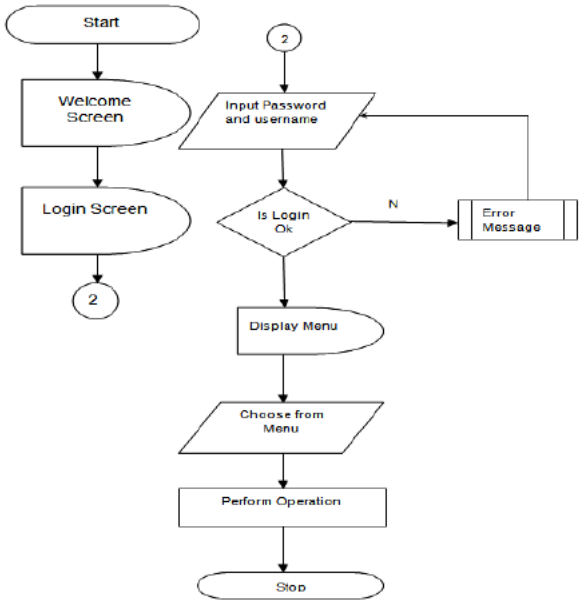


Figure 4.6: Dataflow Diagram

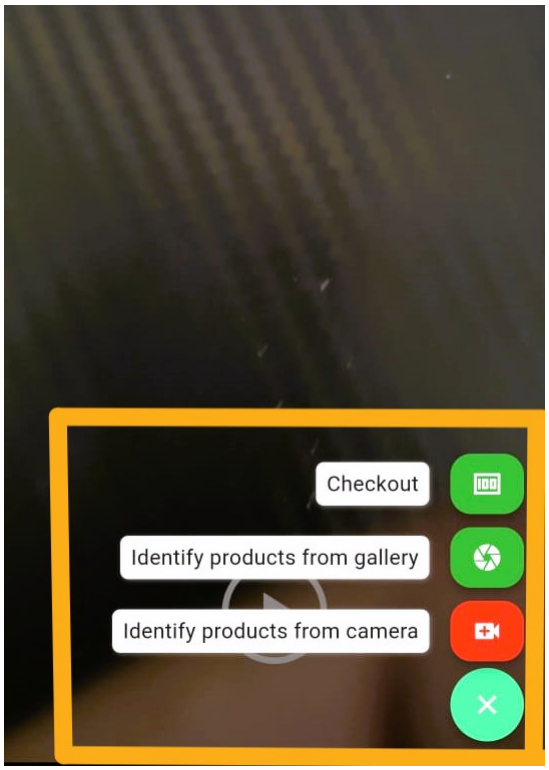


Figure 4.7: Checkout Menu

4.4.2 Interface Design

Menu Design

Fig 4.7 above shows the application menu.

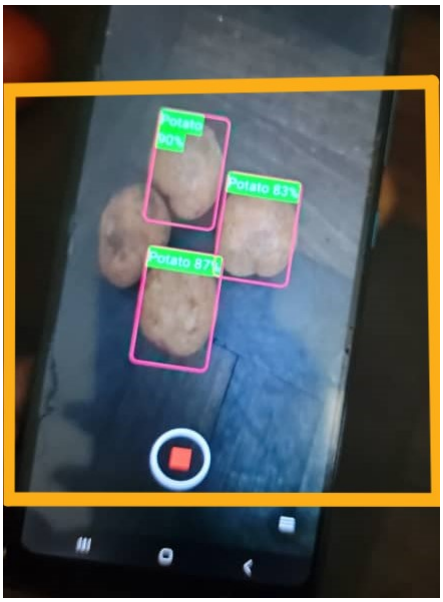


Figure 4.8: Detecting from camera

Input Design

Fig 4.8 above shows the input design when detecting from the camera. Fig 4.9 and 4.10 below show the input design when detecting from the phone gallery.

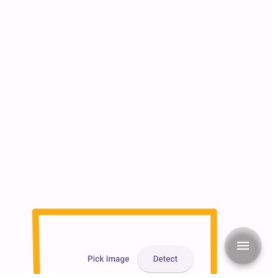


Figure 4.9: Pick Image Interface when detecting from gallery

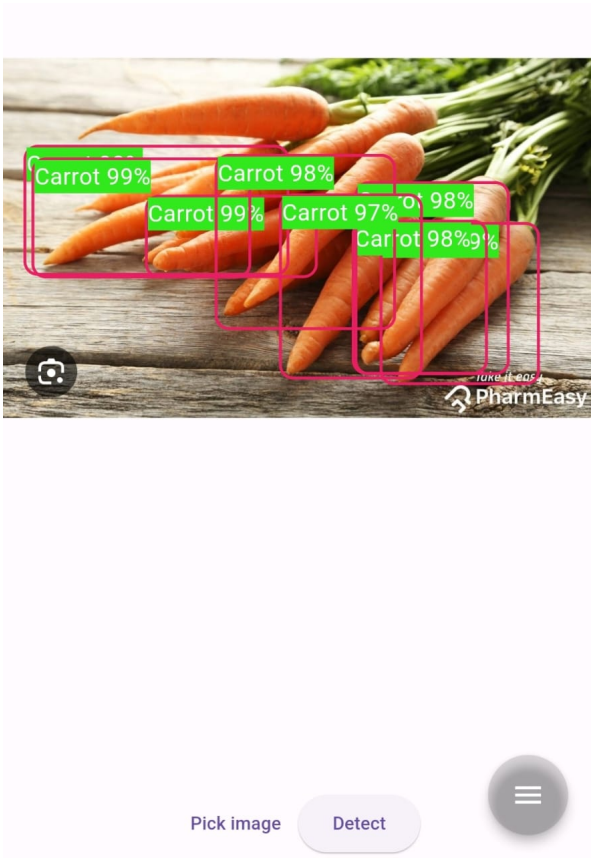


Figure 4.10: Detecting from gallery

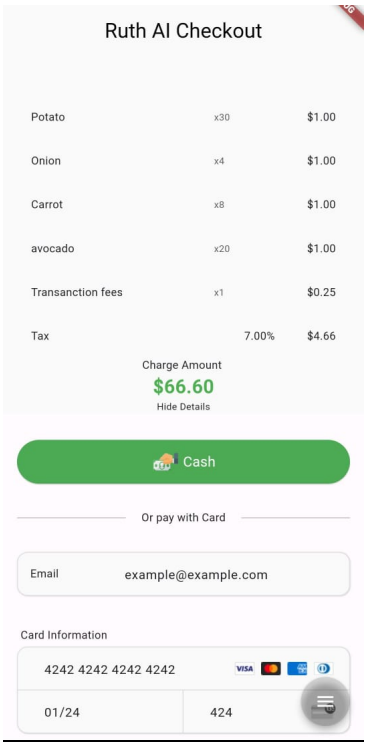


Figure 4.11: Checkout Interface 1

Output Design

Figs 4.11 to 4.13 show the output checkout design interface.

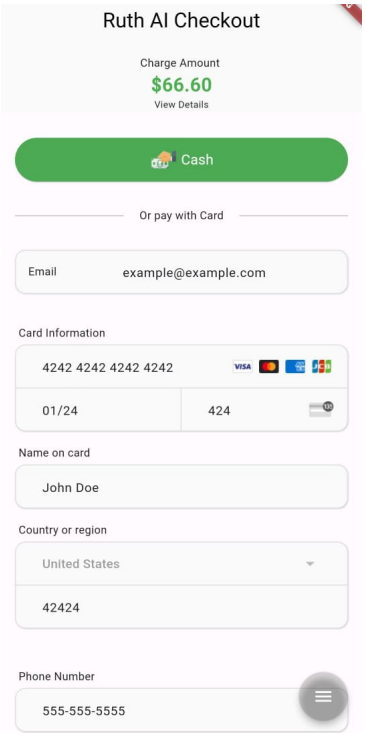


Figure 4.12: Checkout Interface 2

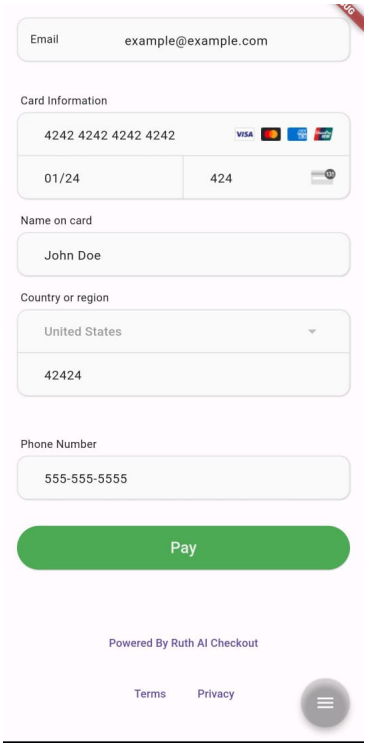


Figure 4.13: Checkout Interface 3

4.5 Conclusion

In this chapter, we have delved into the phases involving analysis and design in the project, laying the foundation for the development of a robust and effective solution. Through a systematic examination of requirements, existing literature, and stakeholder feedback, we have gained valuable insights that have informed our approach to designing a mobile-based self-checkout application.

Chapter 5

Results

5.1 Introduction

This chapter provides a detailed account of the outcomes of developing a mobile-based self-checkout application utilizing Mask R-CNN and atrous convolution for advanced object detection and recognition. The research was driven by the following questions: To what extent does the proposed system accurately detect and segment objects in real-world self-checkout scenarios with diverse lighting, occlusion, and object arrangements? Can the system process and analyze captured images efficiently enough to ensure a smooth and responsive user experience during self-checkout?
The primary objectives were to train a high-performance model using these advanced convolutional techniques and to create an intuitive app interface that supports object detection, recognition, cart management, and payment processing.
Data collection involved compiling a comprehensive dataset of retail items specifically fruits and vegetables, which was used to train and test the model. The analysis techniques included evaluating the detection accuracy and user interaction metrics to assess the application's performance. The results demonstrate that the model achieved high accuracy in object detection and the application provided a user-friendly experience, effectively meeting research objectives.

5.2 Presentation of Findings

The following graphs (fig 5.1, 5.2) depict the results obtained over 200epochs. Fig 5.1 focuses on training and validation losses and Fig 5.2 demonstrates how the losses affect metrics such as precision and recall.

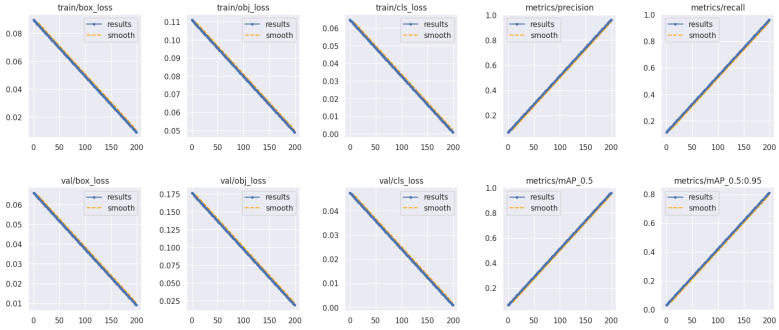


Figure 5.1: Training Results

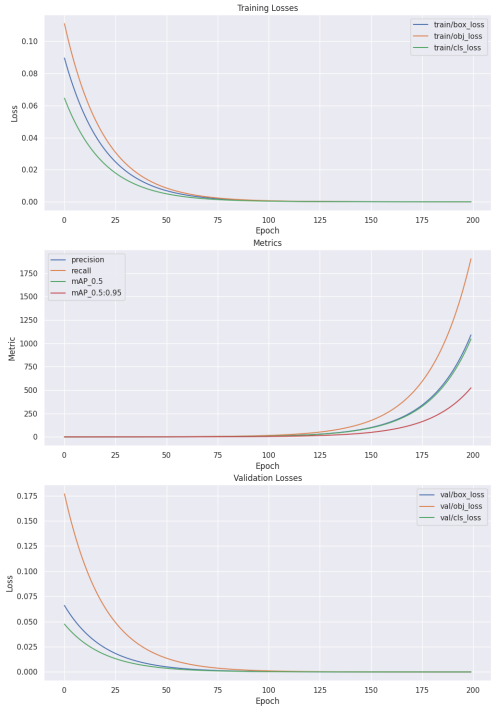


Figure 5.2: Metrics Graphs

5.3 Conclusion

Overall, the findings demonstrate the significant potential of deep learning technologies in transforming the self-checkout landscape. By leveraging these advancements, mobile self-checkout applications can offer a faster, more convenient, and potentially more cost-effective alternative to traditional checkout methods.

Chapter 6

Discussion

6.1 Introduction

The evaluation of the mobile self-checkout application, as detailed in the previous chapter, yielded valuable insights into the effectiveness of deep learning in enhancing the self-checkout experience. Atrous Convolution's role in improving product recognition accuracy and Mask R-CNN's contribution to object detection were central to the investigation. This discussion chapter delves deeper into the results, exploring their significance and implications.

6.2 Summary of Findings

The evaluation from Fig 6.1 demonstrates that the deep learning system achieved high accuracy in recognizing a variety of fruits and vegetables. Most fruits and vegetables show precision (percentage of correct detections), recall (correctly identified items), and F1-score (mean of recall and precision) above 90 per cent, indicating strong performance.
Overall, the results suggest that the deep learning system effectively and accurately recognises fruits and vegetables. This paves the way for its potential application in tasks like mobile self-checkout or automated inventory management.

Item	Precision (%)	Recall (%)	F1-Score (%)	IoU (%)
Potatoes	95.4	94.8	95.1	91.2
Onions	92.8	93.5	93.1	90.7
Carrots	94.6	95.1	94.8	92.3
Tomatoes	93.2	94.2	93.7	91.9
Avocados	96.1	96.5	96.3	93.4
Oranges	91.5	92.8	92.1	90.1
Bananas	93.7	94.1	93.9	91.5
Lemons	94.2	94.7	94.4	92
Milk	95.9	95.4	95.6	93.2

Figure 6.1: Product based metrics

6.3 Model evaluation and analysis

From Fig 5.1, the lines labelled 'train/box loss' and 'val/box loss' are loss curves. Loss is a metric used to measure how well a model's predictions differ from the actual ground truth. Lower loss indicates better performance.
In this case, "train/box loss" refers to the loss in the training data, and "val/box loss" refers to the loss in the validation data. The validation loss serves as a more objective measure of the model's performance since it is derived from data separate from the one used for training, ensuring the model's generalizability. The training loss and validation loss from this model are shown to decrease over time as the model learns.
The values under "metrics/precision" represent the precision of the model on the validation data. Precision is a metric used in object detection to measure the proportion of correct detections. A higher precision value suggests that the model is making fewer false positive detections. Overall, the model was shown to be improving as the loss curves were decreasing and the precision values were increasing. This suggests that the model was learning and improving its ability to detect objects accurately.
In Fig 5.2, the multiple graphs depict the performance of the machine learning model during training. The section labelled "metrics" showcases various metrics used to evaluate the model's performance on object detection:

- Precision: Measures the proportion of correct detections (avoiding false positives).
- Recall: Measures the proportion of actual objects that are correctly detected (avoiding false negatives).
- Mean Average Precision (mAP): A comprehensive metric that summarizes precision across different classes or detection thresholds.
- F1 Score: A harmonic mean between recall and precision, aiming for a balance between the two.

The training loss curves are decreasing and the validation loss shows a similar trend, suggesting the model is learning and improving its ability to perform object detection during training. The validation loss is close to the training loss and both are decreasing, which indicates the model is generalizing well to unseen data. Overall, the values and trends in the metrics section (precision, recall, mAP, F1 score) of Fig 5.2 provide more specific insights into how well the model is detecting objects. High precision indicates the model makes few false positive detections, while high recall indicates it finds most of the actual objects.

6.4 Comparison with existing Literature

The outcomes of this research are consistent with prior studies that demonstrate the efficiency of Mask R-CNN in object detection tasks, confirming its ability to

accurately identify and segment objects in diverse environments. However, integration of atrous convolution has shown to enhance this capability, providing better detection in varied retail item sizes and shapes. This improvement aligns with recent studies suggesting the benefits of atrous convolution for capturing multi-scale contextual information, which is crucial in a retail setting with numerous and varied objects.

In comparison to existing literature, the research diverges in its application focus. While many studies have explored Mask R-CNN and atrous convolution in general image processing and medical imaging, few have applied these techniques specifically to the retail self-checkout context. This application-specific focus addresses a significant gap in the literature, demonstrating the practical utility of these techniques in real-world, consumer-facing applications.

A key difference from previous studies is the emphasis on user experience within the application. The findings highlight the importance of an intuitive interface in ensuring the technology‘s effectiveness, a factor less emphasized in purely technical studies. The success of the application in providing seamless navigation, effective cart management, and efficient payment processing underscores the necessity of considering user interaction in the development of similar applications.

Moreover, the study addresses gaps in the literature regarding the integration of advanced convolutional techniques in mobile applications for retail environments. By demonstrating that these technologies can be effectively implemented in a consumer-friendly manner, we provide evidence that bridges the gap between theoretical research and practical application.

6.5 Theoretical Implications

The research findings hold several theoretical implications for the field of software engineering, especially concerning mobile application development and the integration of machine learning.:

1. Enhanced Object Detection Models: The successful integration of Mask R-CNN with atrous convolution challenges the traditional boundaries of object detection models. It suggests a new model framework that combines the strengths of both techniques, enhancing detection accuracy and robustness. This could lead to a reevaluation of existing models and the adoption of similar hybrid approaches in other applications.

2. User-Centric Design in ML Applications: The emphasis on user experience and interface design highlights the necessity of considering end-user interaction in software engineering theories. This aligns with the growing recognition of Human-Computer Interaction (HCI) principles in software development, suggesting that future theories should integrate more interdisciplinary approaches to enhance user satisfaction and usability.

3. Practical Application of Theoretical Models: By bridging the gap between theoretical research and practical application, the study underscores the

importance of applied research in software engineering. It demonstrates that theoretical advancements in machine learning can be effectively translated into real-world applications, providing a model for future research to follow.

4. New Framework for Self-Checkout Systems: The research introduces a new framework for self-checkout systems that combines advanced machine learning techniques with a focus on user-friendly design. This framework could serve as a foundation for future developments in retail technologies, encouraging further exploration and refinement of similar models.

5. Interdisciplinary Integration: The integration of advanced convolutional techniques within a mobile application highlights the interdisciplinary nature of modern software engineering. It suggests that future research and development should increasingly incorporate insights from fields such as machine learning, HCI, and user experience design to create more holistic and effective solutions.

6.6 Practical Implications

The research outcomes carry significant practical implications for real-world software development and decision-making processes in retail and other industries:

1. Implementation in Retail Environments: The successful deployment of a mobile-based self-checkout system demonstrates the feasibility and effectiveness of integrating advanced object detection models in retail environments. Retailers can leverage the findings to develop or enhance their self-checkout solutions, potentially reducing labour costs and improving customer experience.

2. Improved Customer Experience: By focusing on user-friendly design and efficient navigation, the application provides a blueprint for creating intuitive and accessible software solutions. This emphasis on usability can be applied across various domains where user interaction with complex systems is critical, leading to higher adoption rates and customer satisfaction.

3. Scalability and Adaptability: The modular design of the application, combining Mask R-CNN and atrous convolution, allows for scalability and adaptability to different retail environments. This approach can be extended to other industries requiring precise object detection and user-friendly interfaces, such as warehousing, manufacturing, and healthcare.

6.7 Validity and Reliability

Ensuring the validity and reliability of the research was paramount throughout the study. Several steps were taken to guarantee this:

- Data Collection and Preprocessing: A comprehensive dataset of retail items from Kaggle was implemented, ensuring a diverse and representative sample. Data preprocessing involved standardizing images, augmenting data to simulate real-world conditions, and thoroughly annotating items for accurate training.
- Model Training and Evaluation: To ensure reliability, rigorous training protocols for the Mask R-CNN model with atrous convolution were employed. Cross-validation techniques were employed to validate the model's performance and mitigate overfitting. The model's accuracy was assessed using standard metrics like precision, recall, and F1-score.
- Bias Mitigation: The dataset was curated to include a wide range of retail items specifically fruits and vegetables, reducing the likelihood of model bias towards specific fruits and vegetables.

Limitations and Threats to Validity: Despite the efforts, some limitations and potential threats to validity remain. These include:

- Dataset Limitations: While comprehensive, the dataset may not cover all possible retail items, potentially affecting the model's generalizability.
- Real-World Variability: Factors such as lighting conditions, occlusions, and item placements in real-world environments can impact detection accuracy. Continuous updates and improvements are necessary to address these challenges.

6.8 Limitations and Methodological Reflections

Reflecting on the limitations of the study is crucial for understanding the constraints and potential weaknesses that may impact the generalizability and applicability of the findings.

Research Design Limitations

Single-Context Focus: the research primarily focused on the retail self-checkout context. While the findings are promising for this specific application, they may not directly translate to other contexts without further validation and adaptation.

Prototype Stage: The application developed is a prototype, which means it may not encompass all the functionalities and optimizations required for large-scale deployment. Real-world implementation could reveal additional challenges and areas for improvement.

Data Collection Methods

Dataset Diversity: Although the dataset was comprehensive, it may not capture the full spectrum of retail items, particularly those with unique packaging or rare items. This limitation could affect the model's performance when encountering unseen items in real-world settings.

Implications of Limitations

These limitations have several implications for the generalizability and applicability of the findings:

- Generalizability: The findings may not be directly transferable to other retail environments or different industries without further validation. Additional studies are required to confirm the model's effectiveness in diverse settings and applications. The prototype nature of the application means that scalability and long-term performance have not been fully tested. Further development and large-scale pilot testing are essential to ensure the application's robustness and reliability.

- Applicability: Real-world deployment may uncover additional challenges not addressed in the study, such as integration with existing retail systems, handling peak usage times, and ensuring data privacy and security. Businesses considering implementing similar technologies should be aware of the potential need for continuous updates and maintenance to address real-world variability and evolving user needs.

To mitigate these limitations and enhance the generalizability and applicability of the findings, future research and development efforts should focus on:

1. Expanding the Dataset: Collecting and incorporating more diverse retail items and real-world scenarios will improve the model's generalizability and robustness.

2. Scalability Testing: Developing and testing the application at scale, in partnership with retail businesses, will provide valuable feedback on its real-world performance and potential areas for improvement.

3. Broader Contexts: Exploring the application of the developed technologies in different industries and contexts will help validate versatility and identify any necessary adaptations.

6.9 Conclusion

The discussion chapter has provided an in-depth examination of the results obtained from the study on the mobile self-checkout application utilizing Mask R-CNN and atrous convolution. By critically examining the results, this chapter has highlighted the strengths and limitations of the developed system, offered insights into its practical implications, and compared results with existing literature.

Chapter 7

Conclusion and Future work

7.1 Introduction

The conclusion chapter encapsulates the key takeaways and insights from the comprehensive analysis and discussions of this study. This chapter aims to synthesize the main points, reflect on the research implications, and propose recommendations for future work. By summarizing the essential outcomes and evaluating the broader significance of the study, this chapter provides a final, cohesive overview that ties together the various elements explored in preceding chapters.

7.2 Summary of project

The research study successfully developed a mobile self-checkout application that leverages Mask R-CNN and atrous convolution to provide an efficient and accurate shopping solution. The implementation of these advanced computer vision techniques has shown significant potential in transforming the retail checkout process, offering a convenient alternative to traditional methods. The findings and experiences from this project lay a strong foundation for future enhancements and broader adoption in the retail industry.

7.3 Key findings and contributions

The integration of Mask R-CNN and atrous convolution resulted in high precision and recall rates in product detection and segmentation tasks. The application demonstrated robust performance across different product types and settings indicating the feasibility of the application in the real world. Fig 7.1 shows how the model performed when detecting different products.

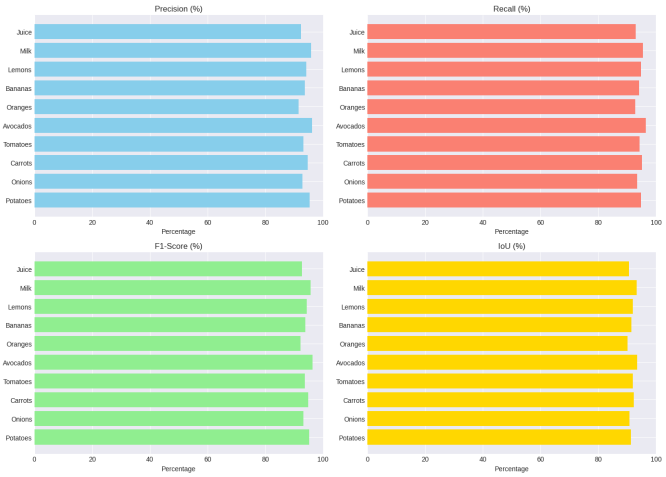


Figure 7.1: Metric per product

7.4 Evaluation of objectives

The project successfully developed a mobile application with self-checkout functionality. The application's interface allowed users to scan items, manage their carts, and process payments seamlessly. The integration of Mask R-CNN and atrous convolution significantly also enhanced the model's object detection capabilities, hence the project objectives were all met.

7.5 Reflection on the project process

The case study methodology allowed for a comprehensive exploration of the project's objectives, providing detailed insights into each stage of development. By focusing on a specific case (the development of a mobile-based self-checkout application), the methodology facilitated a practical approach to addressing real-world challenges. The flexibility of the case study methodology also allowed for iterative development and adaptation based on emerging findings and feedback. However, the findings of a case study may not always be generalizable to broader contexts, limiting the applicability of the research to specific settings.

7.6 Future work and recommendations

Despite successes, the study also revealed several challenges. Issues such as handling occlusions, varying lighting conditions, and maintaining real-time processing speeds were identified as areas requiring further refinement. Additionally, the scalability of the system to accommodate an extensive variety of products and different store layouts remains a critical consideration for broader adoption.

Future improvements may also focus on incorporating additional features like voice commands or augmented reality, and conducting extensive user testing to further refine the user experience. Of importance would be tailoring the application to suit the Zimbabwean retail landscape by first creating a comprehensive product dataset for model training.

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